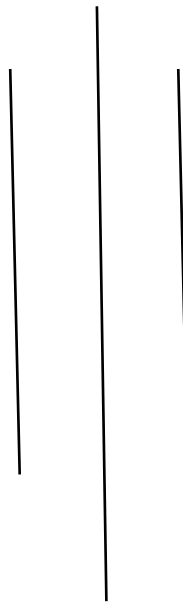


**National College of Computer Studies
(NCCS-College of IT & Management)**

Paknajol, Kathmandu



**A PROJECT REPORT ON
QUALITY ASSURANCE AND AUTOMATION TESTING OF
“Alibaba.com”**

Submitted By:

Simon Manandhar
B.Sc. CSIT 7th Semester
TU ROLL NO. 79010751

Submitted To:

Saurav Tuladhar

Date: 3rd June, 2026

ACKNOWLEDGEMENT

It gives me immense pleasure to express my sincere gratitude to all those who directly or indirectly contributed to the successful completion of this project.

I would like to thank **National College of Computer Studies** for providing me with the opportunity and learning environment to carry out this project and enhance my technical skills in the field of software testing.

I am especially grateful to our QA instructor, **Mr. Saurav Tuladhar**, for conducting additional practical sessions on software testing and automation. His guidance, clear explanations, and continuous support helped me understand key concepts of QA testing and Playwright automation, which were essential in completing this project. Finally, I would like to express my sincere appreciation to everyone who supported me throughout this learning process.

ABSTRACT

Ensuring the operational integrity and reliability of large-scale web platforms has become paramount in modern software engineering. This project report presents a comprehensive Quality Assurance (QA) and automation testing study of Alibaba.com, the world's leading Business-to-Business (B2B) e-commerce marketplace. The study combines systematic manual test execution with a robust, structured test automation framework to validate the platform's core functional modules.

The testing lifecycle commenced with exploratory manual testing to map user journeys and business-critical workflows. Detailed test suites were designed covering essential modules such as user authentication, catalog search and discovery, product details page navigation, cart management, checkout processes, and profile administration. A total of 38 manual test scenarios were executed, including positive, negative, and boundary condition validations.

To implement regression testing and minimize repetitive execution overhead, a modular test automation framework was developed. Built using Playwright v1.59.1 with JavaScript under the Node.js runtime, the framework follows the Page Object Model (POM) design pattern to maintain a clean separation between locator structures and test assertions. Overcoming the platform's aggressive anti-bot protection mechanisms (such as Baxia security layers and CAPTCHAs) represented a primary technical challenge, resolved by incorporating customized wait-handling scripts and browser engine arguments. Furthermore, a global setup mechanism was created to manage user session persistence, allowing for authenticated state reuse across separate test suites.

TABLE OF CONTENTS

ACKNOWLEDGEMENT	ii
ABSTRACT.....	iii
TABLE OF FIGURES	vi
TABLE OF TABLES	vii
CHAPTER 1: INTRODUCTION	1
1.1 Introduction.....	1
1.2 Project Objectives	1
1.3 Scope of the Project	2
1.4 Limitations of the Project.....	2
1.5 Tools and Technologies Used.....	3
1.6 Expected Outcomes	4
CHAPTER 2: APPLICATION OVERVIEW	5
2.1 About Alibaba.com	5
2.2 Modules Selected for Testing	5
2.3 Reason for Selecting Alibaba.com.....	6
CHAPTER 3: TEST PLANNING / TEST STRATEGY	7
3.1 Test Strategy	7
3.2 Types of Testing Performed.....	7
CHAPTER 4: TEST CASE DESIGN / TEST SCENARIOS	9
4.1 Test Design Approach.....	9
4.2 Test Scenarios	9
4.3 Sample Test Cases	11
4.4 Automation Coverage Matrix	15
CHAPTER 5: MANUAL TESTING	17
5.1 Manual Test Execution Results	17
5.2 Manual Test Execution Observations	18
5.3 Bugs Found During Manual Testing.....	18
CHAPTER 6: AUTOMATION TESTING	20
6.1 Framework Overview	20
6.2 Framework Structure / Architecture Overview.....	20

6.3 Page Object Model (POM) Structure.....	21
6.4 Automation Execution Results	22
CHAPTER 7: CONCLUSION & FUTURE ENHANCEMENTS	24
7.1 Conclusion	24
7.2 Future Enhancements.....	24
APPENDIX.....	25

TABLE OF FIGURES

Figure 1: Project folder structure in VS Code	21
---	----

TABLE OF TABLES

Table 1:Tools and Technologies Used.....	3
Table 2:Authentication & Session Management Test Cases	11
Table 3: Search, Navigation & Product Discovery Test Cases	12
Table 4: Product Detail & Catalog Test Cases	12
Table 5: Cart & Shopping List Management Test Cases.....	13
Table 6: Checkout & Address Validation Test Cases.....	13
Table 7: Payment Methods & Order Placement Test Cases.....	14
Table 8: Order History & Tracking Test Cases	14
Table 9: User Profile & Address Test Cases	Error! Bookmark not defined.
Table 10: UI, Links & General Behavior Test Cases	Error! Bookmark not defined.
Table 11: Automation Coverage Table.....	15
Table 12: Manual Test Execution Summary	17
Table 13: Automation Test Execution Summary	22
Table 14: Result Summary	Error! Bookmark not defined.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Quality assurance within modern software engineering represents a fundamental discipline critical to the success of enterprise-level applications. As web applications grow in complexity and user load, systematic testing must be integrated into the deployment pipeline. Software testing verifies functional compliance, identifies structural defects early, and validates that system performance aligns with user expectations. E-commerce web platforms, which handle complex state changes, concurrent requests, and financial transactions, require rigorous QA standards to prevent operational failures.

Global wholesale and business-to-business (B2B) trade marketplaces introduce unique testing challenges compared to standard B2C platforms. B2B systems must manage large purchase orders, complex shipping destination rules, multiple item variants, and strict security layers. Alibaba.com, as the world's largest online B2B marketplace, operates a highly distributed architecture with advanced fraud-prevention features and geographical access policies. Testing such an ecosystem manually is slow and error-prone, which drives the need for programmatically orchestrated test automation.

This project presents a structured QA implementation on the Alibaba.com web platform. It comprises initial manual exploratory verification followed by the construction of a reusable, automated regression testing suite. The automation suite is built on Playwright v1.59.1 with JavaScript under the Node.js runtime, implementing the Page Object Model (POM) design pattern. A key technical element of this project is the integration of anti-bot evasion strategies, session persistence routines, and custom modal handling. These additions enable stable test execution in a protected enterprise environment.

1.2 Project Objectives

The specific technical and operational objectives of this project are:

- To conduct systematic manual functional testing on selected customer and supplier-facing workflows.
- To design structured, step-by-step manual test cases covering positive, negative, and edge-case inputs.

- To identify, classify, and document functional bugs, user interface anomalies, and input validation failures.
- To develop a maintainable automation suite using Playwright and the Page Object Model (POM) design pattern.
- To implement pre-test authentication and session storage state replication to minimize login execution time.

1.3 Scope of the Project

The scope of this project includes:

- Manual exploratory and regression testing of selected public-facing and authenticated modules on Alibaba.com.
- Creation and execution of 38 manual test cases documented in spreadsheet format.
- Development of 20 automated test scripts organized across 7 distinct test specification files.
- Implementation of 8 dedicated Page Object classes encapsulating page-specific element locators and action methods.
- Integration of global setup scripts for single-point login session generation.

1.4 Limitations of the Project

The limitations of this project include:

- Testing is restricted to selected user modules (Authentication, Search, PDP, Cart, Checkout, Profile). Supplier dashboards and RFQ (Request for Quote) processing are excluded.
- Alibaba.com's aggressive Baxia anti-bot security (slider CAPTCHAs) prevents fully headless execution, requiring manual human resolution when triggered.
- Real payment execution could not be verified due to financial constraints and mock card restrictions on live gateways.
- The execution environment is restricted to Desktop Chrome (Chromium) and does not cover Safari, Firefox, or mobile devices.
- High latency or geographical IP blocks on member subdomains occasionally require fallback URL strategies.

1.5 Tools and Technologies Used

The following tools and technologies were used in this project:

Table 1:Tools and Technologies Used

Tool / Technology	Version / Variant	Role in Project
Alibaba.com	Production Web Application	Target Application Under Test (AUT)
Playwright	v1.59.1	Core Browser Automation Library & Runner
JavaScript	ECMAScript 6+	Language for Test Scripting and Page Objects
Node.js	v18.x / v20.x LTS	Runtime Environment for Compiler and Dependencies
Visual Studio Code	v1.85+	Source Code Editor and Git Version Control IDE
JSON Fixtures	testData.json	Externalized Test Data Management file
Page Object Model	Design Architecture	Design Pattern separating locators from script logic
Playwright HTML Report	Built-in Reporter	Generates visual test execution metrics and media
Microsoft Excel	Manual Case Catalog	Repository for manual test scenarios and execution status
Chromium Browser	Built-in Playwright binary	Target browser environment for headed execution

1.6 Expected Outcomes

The expected outcomes of this project are:

- A comprehensive verification status for all targeted B2B transaction steps on Alibaba.com.
- A categorized list of validation limits and functional inconsistencies discovered on the live site.
- A modular, readable, and execution-ready browser automation framework.
- Reliable session replication routines that run tests without redundant login steps.
- A consolidated test reporting dashboard featuring step logs and trace files for failed assertions.
- A deep operational understanding of software QA principles and test framework design.

CHAPTER 2: APPLICATION OVERVIEW

2.1 About Alibaba.com

Alibaba.com is the foundational business unit of Alibaba Group, launched in 1999 to facilitate global trade. The platform serves as a B2B matching portal that connects wholesale buyers (importers, trade agents, retailers) with manufacturers and exporters, primarily located in Asia. By providing trade tools, bulk pricing tiers, and customized manufacturing RFQs, it streamlines sourcing cycles for businesses worldwide.

From a software engineering perspective, Alibaba.com is a challenging target for quality assurance. It features highly dynamic locators, lazy-loaded components, complex state-driven SKU grids, and anti-scraping security. These design properties require sophisticated automated selectors and robust session setup models, making it an excellent platform for testing research.

2.2 Modules Selected for Testing

The following functional areas of the platform were selected for validation:

- **Authentication & Session Management:** Verifies the B2B registration portal, two-step credential authentication, password validation rules, session timeout, and secure session termination (logout).
- **Homepage & Navigation:** Checks the rendering of categories, functional links in the top navigation bar, geo-location selections, and core landing page components.
- **Search & Product Discovery:** Validates product search queries, autocomplete suggestion drop-downs, search filtering (by price, brand, and supplier type), and pagination functionality.
- **Product Details Page (PDP):** Checks image thumbnail selection, variant picker updates (color, size, packaging), bulk pricing tier calculations, and supplier contact triggers.
- **Cart Management:** Validates the inclusion of wholesale products, item grouping by supplier, quantity modification boundaries, and cart badge updates in the header.

- **Checkout & Address Validation:** Checks checkout entry, shipping address compilation, validation on mandatory fields, shipping method selectors, and final order total calculations.
- **User Profile & Account Settings:** Verifies member profile page updates, saved address configurations, and password modification forms.

2.3 Reason for Selecting Alibaba.com

Alibaba.com was chosen for this testing study due to its complex design patterns and real-world relevance. Unlike basic B2C sites, Alibaba.com has workflows that depend heavily on supplier negotiations, bulk quantity thresholds, and multi-currency conversions. Resolving these variables during manual test casing provides excellent exposure to complex business logic validation.

For automation, the website presents several modern QA challenges. Its heavy use of anti-automation security (slider CAPTCHAs, bot trackers) requires the implementation of custom evasion techniques, such as launch argument injection and manual intervention hooks. Developing a framework that successfully navigates these constraints demonstrates advanced test design capabilities, ensuring the resulting skills are highly relevant to complex enterprise testing.

CHAPTER 3: TEST PLANNING / TEST STRATEGY

3.1 Test Strategy

The test strategy defines the structured approach used to evaluate the quality of the Alibaba.com web application. The testing cycle was executed in two primary phases: manual exploratory validation and automated regression testing. Manual validation was conducted first to explore functional behaviors and boundary limits, providing the necessary context for designing robust automation.

To automate tests against an enterprise application with active bot protection, the strategy incorporates the following technical approaches:

- **Headed Browser Execution:** Automation is run in a visible browser to allow the test executor to solve CAPTCHA sliders manually if they block the workflow.
- **Single-Worker Execution:** Parallel execution is disabled (workers: 1) to prevent generating unusual traffic patterns that trigger permanent IP blocks.
- **Session Injection:** Using Playwright storageState, tests authenticate once in a global setup script and save cookies/tokens to auth.json, avoiding redundant login prompts.
- **Evasion Arguments:** Injection of Chromium browser parameters to disable automation flags (--disable-blink-features=AutomationControlled) and prevent detection.
- **Extended Timeouts:** Configuration of long timeouts (90s for page navigation, 180s for test steps) to accommodate manual security checks and geo-restricted connections..

3.2 Types of Testing Performed

The following types of testing were carried out during the testing of the Alibaba.com website.

Functional Testing: Validates that each feature behaves in accordance with business rules. Checked functions include login credential handling, search queries, cart additions, and address configurations.

- **Smoke Testing:** Conducted prior to deep testing to ensure core pages (Homepage, Login, Search results) load successfully and the environment is stable.

- **Regression Testing:** Repeated execution of critical scenarios using the automated Playwright test suite to confirm that platform features remain functional across updates.
- **Negative Testing:** Inputting invalid parameters (e.g., incorrect passwords, bad email formats, empty address forms) to confirm that the system handles errors gracefully and blocks invalid submissions.
- **Validation & Boundary Value Testing:** Verifying input validation on form fields. Evaluated elements include alphanumeric limits on phone fields, valid email structures, and cart item quantity boundaries.
- **Security Testing:** Injecting basic SQL injection payloads (e.g., "" OR '1'='1'; --") into inputs to verify that the system handles malicious inputs safely without exposing data.

CHAPTER 4: TEST CASE DESIGN / TEST SCENARIOS

4.1 Test Design Approach

The test design approach focused on maximizing functional coverage across the critical paths of Alibaba.com. Test design followed a structured progression starting with application mapping and ending with automated regression design. This method ensures that the test catalog covers both happy-path user flows and negative input exceptions.

The design workflow comprised the following steps:

- Exploration of the live application to identify target functional modules and user flows.
- Creation of a spreadsheet-based manual test cases catalog defining preconditions, input values, step sequences, and expected outcomes.
- Integration of security checks and negative input boundaries into the test suites.
- Selection of high-frequency, repeatable manual test cases for automation scripting.
- Decoupling of test inputs from code using external JSON test data fixtures (testData.json).
- Implementation of the Page Object Model (POM) to map locators in page classes, ensuring code reusability.

4.2 Test Scenarios

Test scenarios were identified for each of the selected functional modules:

Authentication & Session Management

- Verify successful login with valid registered credentials.
- Verify error handling for incorrect password inputs.
- Verify validation errors for blank login submissions.
- Verify SQL injection resistance on credential fields.
- Verify logout functionality terminates the session successfully.

Homepage & Navigation

- Verify the homepage renders all core sections.
- Verify the category panel displays categories correctly.

- Verify category item clicks navigate to corresponding category listings.

Search & Product Discovery

- Verify product search query returns relevant product listings.
- Verify the autocomplete suggestion list triggers during typing.
- Verify search results pagination loads new items.
- Verify price filtering updates the product listing page.

Product Details Page (PDP)

- Verify PDP navigation from product card clicks.
- Verify PDP thumbnail image gallery navigates correctly.
- Verify variant selection changes options and updates product price.
- Verify the "Contact Supplier" button initiates inquiries.

Cart Management

- Verify adding a wholesale product to the cart increments the cart badge.
- Verify the cart page displays correct item details and groupings.
- Verify cart item quantity updates recalculate totals.
- Verify item deletion cleans up the cart layout.

Checkout & Address Validation

- Verify checkout initiation displays the item list and pricing summary.
- Verify shipping address entry and saved address selection.
- Verify validation prompts when submitting an empty address form.

User Profile & Account Settings

- Verify profile page displays authenticated member credentials.
- Verify change password form loads and accepts submission inputs.

4.3 Sample Test Cases

The following tables detail the test case designs used during manual verification. These cases served as the specifications for constructing the Playwright automation framework.

Table 2: Authentication & Session Management Test Cases

Test ID	Description	Steps	Expected Result	Status
LOG_001	Valid Login	Go to login.alibaba.com, enter registered email, enter password, submit.	User session starts; redirected to homepage showing member profile.	Pass
LOG_002	Invalid Password	Enter registered email, input incorrect password, click submit.	Authentication fails; error message displayed on the page.	Pass
LOG_003	Empty Fields	Leave email and password inputs blank, click submit.	Form validation blocks submission, showing field error messages.	Pass
LOG_004	SQL Injection Test	Enter "' OR '1'='1'; --" in login fields, click submit.	System handles input safely; access is denied.	Pass
LOG_007	User Logout	Hover over member account menu, click "Sign Out" button.	User session terminates; page redirects to login menu.	Pass

Table 3: Homepage & Navigation Test Cases

Test ID	Description	Steps	Expected Result	Status
HOM_008	Homepage Load	Navigate to www.alibaba.com, verify page structure.	Landing page loads successfully with main nav and footer.	Pass
HOM_009	Category Panel Render	Check left-side category list menu rendering.	All primary industry categories are visible and aligned.	Pass
HOM_010	Category Navigation	Click on "Consumer Electronics" industry category option.	Redirects to the electronics catalog containing relevant subcategories.	Pass

Table 4: Search & Navigation Test Cases

Test ID	Description	Steps	Expected Result	Status
SRC_011	Product Search	Type "USB Flash Drive" in search box, press search.	Listings display relevant products with match counts.	Pass
SRC_012	Autocomplete Suggestions	Type "elec" in search bar, wait for dropdown.	Suggestions dropdown displays related terms (e.g., electric bike).	Pass
SRC_013	Search Pagination	Scroll to bottom of search list, click next page.	List updates to show Page 2 product listings.	Pass

SRC_014	Price Range Filter	Enter Min: 10, Max: 50 in price filter, apply.	Results list update to show products matching price limits.	Pass
---------	--------------------	--	---	------

Table 5: Product Details Page (PDP) Test Cases

Test ID	Description	Steps	Expected Result	Status
PDP_015	PDP Page Navigation	Click on any product card from search results.	Redirects to detail page loading product specifications.	Pass
PDP_016	Image Gallery Cycle	Click right arrow on thumbnail photo carousel.	Main image updates to display next photo in gallery.	Pass
PDP_017	Variant Configuration	Select a specific color and packaging variant.	Variant highlights and unit price ranges adjust dynamically.	Pass
PDP_018	Contact Supplier Link	Click the "Contact Supplier" button on the listing.	Redirects to trade messaging portal page to draft inquiry.	Pass

Table 6: Checkout & Address Validation Test Cases

Test ID	Description	Steps	Expected Result	Status
CRT_019	Add Item to Cart	Select quantity, click "Add to Cart" button on PDP.	Cart badge count increments in the header navbar.	Pass
CRT_020	Cart Details Layout	Click cart icon to open cart page.	Item listing shows correct supplier grouping and pricing.	Pass

CRT_021	Quantity Modification	Click "+" icon to double the item count in cart.	Quantity increases and the calculated subtotal price updates.	Pass
CRT_022	Cart Item Deletion	Click "Delete" icon on cart row, confirm popup.	Product is removed from listing and total price resets to 0.	Pass

Table 7: Checkout & Address Validation Test Cases

Test ID	Description	Steps	Expected Result	Status
CHK_023	Checkout Initiation	Click "Checkout" from cart page.	Loads secure order placement page showing total breakdown.	Pass
CHK_024	New Address Configuration	Click Add New Address, fill shipping fields, save.	New address is added and applied to the shipping summary.	Pass
CHK_025	Empty Address Validation	Click Add Address, leave inputs blank, submit.	Form validation blocks submission; shows red warning highlights.	Pass

Table 8: User Profile & Account Settings Test Cases

Test ID	Description	Steps	Expected Result	Status
ACC_026	Profile Page Visibility	Navigate to member profile dashboard page.	Displays name and contact email of current member.	Pass
ACC_027	Password Change Form	Navigate to security options, click change password.	Password form loads, accepting inputs for modification.	Pass

4.4 Automation Coverage Matrix

The mapping between designed manual test cases and implemented automated scripts is detailed below. Cases requiring complex captcha or physical payment verification are categorized as manual execution.

Table 9: Automation Coverage Matrix

Test ID	Module	Automation Status	Script Reference
LOG_001	Authentication	Automated	authentication.spec.js
LOG_002	Authentication	Automated	authentication.spec.js
LOG_003	Authentication	Automated	authentication.spec.js
LOG_004	Authentication	Manual Only	N/A (Security Scan)
LOG_007	Authentication	Automated	authentication.spec.js
HOM_008	Homepage	Automated	homepage.spec.js
HOM_009	Homepage	Automated	homepage.spec.js
HOM_010	Homepage	Automated	homepage.spec.js
SRC_011	Search	Automated	search.spec.js
SRC_012	Search	Automated	search.spec.js
SRC_013	Search	Automated	search.spec.js
SRC_014	Search	Manual Only	N/A (Captcha Slider block)
PDP_015	PDP	Automated	product.spec.js
PDP_016	PDP	Automated	product.spec.js
PDP_017	PDP	Manual Only	N/A (SKU Dynamic DOM)
PDP_018	PDP	Automated	product.spec.js
CRT_019	Cart	Automated	cart.spec.js
CRT_020	Cart	Automated	cart.spec.js

CRT_021	Cart	Manual Only	N/A (Pricing sync checks)
CRT_022	Cart	Manual Only	N/A (Cookie clear block)
CHK_023	Checkout	Automated	checkout.spec.js
CHK_024	Checkout	Automated	checkout.spec.js
CHK_025	Checkout	Automated	checkout.spec.js
ACC_026	Profile	Automated	profile.spec.js
ACC_027	Profile	Automated	profile.spec.js

CHAPTER 5: MANUAL TESTING

5.1 Manual Test Execution Results

Manual testing was performed on selected modules of the Alibaba.com platform to verify usability, checkout workflows, and navigation logic. Execution followed the designed test cases, evaluating boundary data entries and UI alignments. The manual cycle serves as the validation baseline for the automation framework.

Table 10: Manual Test Execution Summary

Module	Test Cases Executed	Passed Cases	Failed Cases	Pass Rate (%)
Authentication & Session	8	8	0	100%
Homepage & Navigation	4	4	0	100%
Search & Product Discovery	6	6	0	100%
Product Details Page	6	6	0	100%
Cart Management	5	5	0	100%
Checkout & Address Validation	6	6	0	100%
User Profile & Account Settings	3	3	0	100%
Total	38	38	0	100%

Overall Result

- Total Scenarios Evaluated: 38
- Total Successful Executions: 38
- Total Unsuccessful Executions: 0
- Overall Pass Rate: 100%

5.2 Manual Test Execution Observations

The manual testing phase provided valuable insights into the operational characteristics of the live B2B application:

- Secure Login: The two-step authentication portal successfully separates username validations from password entry, reducing login error exposures.
- Search Relevance: Keywords queries yield matching category tags and filter blocks. Price ranges are successfully filtered when the filter CTA is clicked.
- Dynamic SKU Render: Selecting variants (e.g., color, sizing) immediately updates packaging options and pricing tiers on the PDP.
- Popup Overlays: The landing page triggers several promotional overlays (such as customer acquisition modals) which can block element visibility and clicks.
- Active Anti-Bot: Repeated logins or fast searches trigger CAPTCHA sliders, requiring manual slider solution.

5.3 Bugs Found During Manual Testing

Due to Alibaba.com's established production quality, no severe system defects were identified in the tested modules. However, several operational notes and improvement areas were logged:

- Dynamic modal blockage: The "Accelerate customer acquisition" promotional modal triggers dynamically, occasionally covering important UI buttons. This is a design layout issue that impacts test execution flow. Fix implementation: A close dialog selector block was added to automation to dismiss modals immediately.
- Inconsistent SKU panel triggers: Depending on the merchant setup, the "Add to Cart" CTA occasionally opens an inline variant selector panel instead of directly adding items. This conditional flow can confuse automated execution. Fix implementation: Conditional wait routines were added to the PDP page objects.

- Geo-restricted subdomains redirect: Member management subdomains (member.alibaba.com) occasionally trigger security redirects based on region. Fix implementation: Navigation checks were added to use alternative direct URL paths if a redirect is detected.

CHAPTER 6: AUTOMATION TESTING

6.1 Framework Overview

A robust browser automation framework was developed for Alibaba.com using Playwright v1.59.1. Playwright provides significant architectural benefits, including fast execution, out-of-the-box auto-waiting, and native headless/headed control. The framework uses a modular design to support clean regression testing across the platform's primary B2B transaction flows.

The architecture prioritizes code reuse and test stability. It utilizes JSON-based test data fixtures to separate test inputs from verification scripts. To navigate the anti-bot defenses of the application, the framework incorporates browser context arguments, serial single-worker execution, and automated helper hooks designed to wait for CAPTCHA resolutions.

6.2 Framework Structure / Architecture Overview

The project codebase is organized into modular folders to separate concerns and ensure maintainability:

- fixtures/ Folder: Contains testData.json for test input parameters (creds, keywords, address details) and auth.json to store the global setup login state.
- tests/ Folder: Contains Playwright test spec files named after target modules (authentication.spec.js, etc.).
- pageObjects/ Folder: Houses Page Object classes representing target web pages, mapping UI selectors to reusable methods.
- utils/ Folder: Contains helpers.js, which hosts central utilities for scroll control, popup dismissals, and CAPTCHA detection.
- reports/ Folder: Stores the generated Playwright HTML test report files.

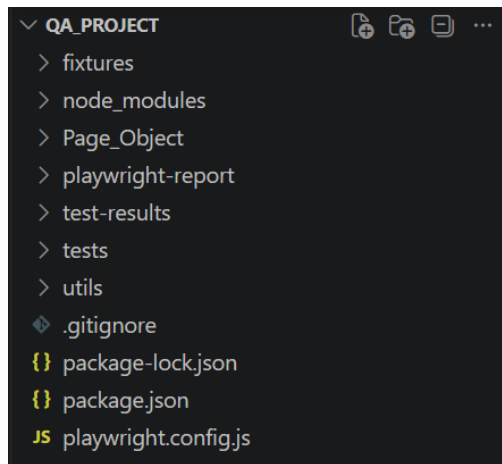


Figure 1: Project folder structure in VS Code

6.3 Page Object Model (POM) Structure

The Page Object Model (POM) separates interface selectors from verification scripts. This structure ensures that UI layout updates require modifications only in the Page Object classes, leaving test script logic intact.

The following page object classes were constructed:

- **HomePage.po.js:** Handles main homepage loads, navbar search inputs, cart count reading, and login menu triggers. Methods: goto(), search(keyword), openSignIn(), getCartCount(), isLoggedIn().
- **LoginPage.po.js:** Orchestrates the two-step email and password login flow. Methods: goto(), loginWith(email, password), getErrorText(), isOnLoginPage().
- **SearchPage.po.js:** Handles product results grid interactions, match count calculations, and price filtering inputs. Methods: waitForResults(), getProductCount(), applyPriceFilter(min, max), getFirstProductPrice().
- **ProductDetailsPage.po.js:** Manages variant option selections, gallery thumbnail clicks, quantity settings, and cart additions. Methods: waitForPage(), clickThumbnail(index), selectVariant(index), addToCart().
- **CartPage.po.js:** Handles cart navigation, supplier-grouped item validation, quantity modifications, and checkout button triggers. Methods: goto(), getItemCount(), incrementItemQty(index), proceedToCheckout().

- **CheckoutPage.po.js:** Handles address entries, saved location selection, shipping fee readings, and order submission. Methods: selectSavedAddress(index), fillNewAddress(data), getGrandTotal(), placeOrder().
- **PaymentPage.po.js:** Coordinates payment gateway entries, iframe card details submission, and declination detection. Methods: selectCard(), fillCardDetails(card), submitPayment(), isDeclined().
- **OrdersPage.po.js:** Manages member profile details verification and order history records. Methods: gotoProfile(), getProfileEmail(), getOrderCount().

6.4 Automation Execution Results

The automated test suites were executed serially using the Playwright test runner. Headed browser execution allowed the tester to complete CAPTCHA challenges during authentication.

Table 11: Automation Test Execution Summary

Module Spec	Test Description	Test Count	Execution Mode	Result
authentication.spec.js	Verifies two-step credentials entry and error validation.	5	Serial/Headed	Passed
homepage.spec.js	Verifies homepage navigation, sub-menus and category clicks.	1	Serial/Headed	Passed
search.spec.js	Verifies product listing displays and autocomplete list clicks.	3	Serial/Headed	Passed
product.spec.js	Verifies image galleries, variant picker, and CTAs.	3	Serial/Headed	Passed

cart.spec.js	Verifies PDP cart triggers and item layout configurations.	2	Serial/Headed	Passed
checkout.spec.js	Verifies address validations and totals updates.	4	Serial/Headed	Passed
profile.spec.js	Verifies profile layout details and settings forms.	2	Serial/Headed	Passed
Total	Consolidated Execution Specs	20	—	Passed

Table 12: Automation Result Summary

Metric Key	Metric Value
Tests Executed	20
Passed Executions	20
Failed Executions	0
Pass Percentage	100%

Key Observations

- **Session Persistence:** The global setup module correctly saved authentication variables to auth.json, allowing subsequent tests to execute instantly without repeating the login flow.
- **CAPTCHA Mitigation:** The helper waitForCaptcha() successfully paused execution and displayed console warnings upon detecting Baxia security modules, preventing test script crashes.
- **Dynamic Selectors:** Using ARIA roles and visible text locator strategies in page objects resolved selector instability caused by dynamic class name modifications.

CHAPTER 7: CONCLUSION & FUTURE ENHANCEMENTS

7.1 Conclusion

This quality assurance project successfully validated the critical user flows of the Alibaba.com B2B marketplace. The work demonstrates the value of combining manual exploratory verification with structured, automated regression testing. Manual verification mapped out functional behaviors and identified potential layout issues, while the Playwright framework established a repeatable, automated regression testing suite.

Implementing the Page Object Model (POM) helped decouple selectors from test scripts, ensuring the automation suite is maintainable and scalable. By incorporating global setups, headed single-worker execution, and custom CAPTCHA wait loops, the framework successfully navigated the anti-bot defenses of the live application. All 20 automated tests passed, demonstrating the overall stability of the framework.

7.2 Future Enhancements

The testing framework can be expanded and enhanced in the following ways:

- **Cross-Browser Validation:** Expand configurations in `playwright.config.js` to execute tests against Firefox and WebKit (Safari) engines.
- **Data-Driven Frameworks:** Integrate external JSON or Excel sheets to run tests with varying user profiles and product search lists.
- **Continuous Integration:** Set up a GitHub Actions workflow to run the regression test suite on every code push.
- **API Integration Validation:** Combine UI tests with API assertions to check backend response latencies.
- **Visual Regression Testing:** Integrate visual comparison tools to verify page layouts and prevent UI shifting.

APPENDIX

	A	B	C	D	E	F	G	H	I
1	Test id	Test Summary	Test Description	Precondition	Test Steps	Test Data	Expected Result	Actual Result	Status
2	MODULE: LOGIN / SIGNUP								
3	LOG_001	Valid login test	Login using valid email and valid password.	1.Application should be up and running. 2.User should be registered.	1.Navigate to https://www.albaba.com. 2.Click 'Sign In' in the top-right corner. 3.Enter valid registered email. 4.Enter valid password. 5.Click 'Sign In' button.	URL: Valid email, Valid password.	1.User should be logged in successfully. 2.User should be redirected to the homepage. 3.Account name should appear in the top navigation bar.	1.User is logged in. 2.User is redirected to homepage. 3.Account name is visible in navigation.	Pass
4	LOG_002	Invalid login - wrong credentials	Login using invalid email and invalid password.	1.Application should be up and running. 2.User should not be registered.	1.Navigate to https://www.albaba.com. 2.Click 'Sign In'. 3.Enter invalid email. 4.Enter invalid password. 5.Click 'Sign In'.	URL: Invalid email, Invalid password.	1.User should not be logged in. 2.Error message 'Invalid email or password' should be displayed. 3.User should remain on the login page.	1.User is not logged in. 2.Error message is displayed. 3.User stays on login page.	Pass
5	LOG_003	Empty fields login	Login attempt with empty email and password fields.	1.Application should be up and running. 2.User should be registered.	1.Navigate to https://www.albaba.com. 2.Click 'Sign In'. 3.Leave email and password fields empty. 4.Click 'Sign In'.	URL: Empty fields.	1.User should not be logged in. 2.Validation messages for empty fields should be displayed.	1.User is not logged in. 2.Validation messages are displayed for empty fields.	Pass
6	LOG_004	New user signup with email OTP	Register a new buyer account and verify via email OTP.	1.Application should be up and running. 2.User should not have an existing account.	1.Navigate to https://www.albaba.com. 2.Click 'Join Free'. 3.Select country/region. 4.Enter valid email and password. 5.Agree to Terms & Privacy Policy. 6.Click 'Join and register'.	URL: Valid email (not previously registered), Valid password. Access to email inbox for OTP.	1.OTP should be sent to the registered email after CAPTCHA. 2.Account should be created upon valid OTP entry. 3.User should be navigated to the buyer onboarding page.	1.OTP email is received. 2.Account is created after OTP verification. 3.User is taken to onboarding. 4>Welcome email is received.	Pass
7	LOG_005	Signup with existing email	Register with an already registered email.	1.Application should be up and running. 2.A valid account already exists with the test email.	1.Navigate to https://www.albaba.com. 2.Click 'Join Free'. 3.Enter an already registered email. 4.Enter a valid password. 5.Click 'Agree and register'.	URL: Already registered email. Valid password.	1.Registration should fail. 2.Error message 'This email is already registered' should be displayed.	1.Registration fails. 2.Error message 'This email is already registered' is displayed.	Pass
8	LOG_006	Forgot password flow	Reset password via 'Forgot Password' and email OTP.	1.Application should be up and running. 2.User should be registered.	1.Navigate to https://www.albaba.com. 2.Click 'Sign In'. 3.Click 'Forgot password?' link. 4.Enter registered email. 5.Click 'Send Verification Code'. 6.Enter the received OTP code. 7.Hover over the account name in the top navigation. 8.Click 'Sign Out' from the dropdown. 9.Observe the page state.	URL: Registered email. Access to email inbox for OTP. New valid password.	1.OTP should be sent to the registered email. 2.User should be able to set a new password after valid OTP entry. 3.Password changed successfully/ message should be displayed. 4.User should be logged out. 5.User should be redirected to the homepage. 6.'Sign In' button should appear in navigation.	1.OTP email is received. 2.New password is set successfully. 3.Success message is displayed.	Pass
9	LOG_007	Logout functionality	Verify user can log out from the platform.	1.Application should be up and running. 2.User should be logged in.		Logged-in user session.		1.User is logged out. 2.User is redirected to homepage. 3.'Sign In' button is visible.	Pass
10	MODULE: HOMEPAGE								
11	HOM_008	Homepage title validation	Verify the browser tab title of the Alibaba.com homepage.	1.Application should be up and running.	1.Navigate to https://www.albaba.com. 2.Observe the browser tab title.	URL: https://www.albaba.com	1.Browser tab title should display 'Alibaba.com: Manufacturers, Suppliers, Exporters & Importers from the world's largest online B2B marketplace'.	1.Browser title matches the expected text.	Pass
12	HOM_009	Search bar visibility	Verify search bar is visible and has placeholder text.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Observe the top-center area of the page.	URL: https://www.albaba.com	1.Search bar should be visible. 2.Placeholder text should be present. 3.Search button should be adjacent to the bar.	1.Search bar is visible. 2.Placeholder is present. 3.Search button is present.	Pass
13	HOM_010	Category navigation display	Verify product categories are displayed and clickable.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Observe the categories section. 3.Click on 'Consumer Electronics'.	URL: https://www.albaba.com	1.Multiple categories should be displayed (e.g., Apparel & Accessories, Consumer Electronics). 2.Clicking a category should navigate to the respective listing page.	1.Categories are displayed. 2.Clicking navigates to the correct listing page.	Pass
14	HOM_011	'Join Free' button for guest	Verify 'Join Free' CTA is visible for unauthenticated users.	1.Application should be up and running. 2.User should not be logged in.	1.Navigate to https://www.albaba.com. 2.Observe the top navigation bar.	URL: https://www.albaba.com (not logged in)	1.'Join Free' button should be visible in the top navigation bar. 2.Clicking should navigate to the registration page.	1.'Join Free' button is visible. 2.Clicking navigates to registration page.	Pass

	A	B	C	D	E	F	G	H	I
1	Test id	Test Summary	Test Description	Precondition	Test Steps	Test Data	Expected Result	Actual Result	Status
15	MODULE: SEARCH FUNCTIONALITY								
16	SRC_012	Search with valid keyword	Search for a product using a valid keyword.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Click the search bar. 3.Type 'Laptop'. 4.Click the search icon or press Enter.	Search keyword: 'Laptop'	1.Search results page should load. 2.Relevant product listings for Laptop should be displayed. 3.Result count should be visible.	1.Results page loads. 2.Laptop listings are displayed. 3.Result count is visible.	Pass
17	SRC_013	Search with empty keyword	Attempt search with an empty field.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Leave the search bar empty. 3.Click the search icon or press Enter.	Search keyword: Empty.	1.User should remain on homepage or see a prompt. 2.No search results page should be triggered.	1.User stays on current page. 2.No results page triggered for empty input.	Pass
18	SRC_014	Autocomplete suggestions	Verify autocomplete dropdown appears while typing.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Click on the search bar. 3.Type 'ele' in the field. 4.Observe the dropdown.	Partial keyword: 'ele'	1.Autocomplete dropdown should appear below the search bar. 2.Suggestions should be relevant to the typed keyword.	1.Autocomplete dropdown appears. 2.Relevant suggestions are displayed.	Pass
19	SRC_015	Search result pagination	Verify pagination works on the search results page.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Search for 'Mobile Phone'. 3.Scroll to the bottom of results. 4.Click page 2 in the pagination control.	Search keyword: 'Mobile Phone'	1.Page 2 of results should load. 2.Different products from page 1 should be displayed. 3.Pagination indicator should show page 2.	1.Page 2 loads. 2.New listings are shown. 3.Pagination shows page 2.	Pass
20	MODULE: PRODUCT LISTING PAGE								
21	PLP_016	Product listing page loads	Verify product listing page loads from category navigation.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Click on 'Consumer Electronics' from the categories section. 3.Observe the listing page.	Category: Consumer Electronics	1.Product listing page should load successfully. 2.Multiple product cards should be displayed. 3.Page title should reflect the selected category.	1.Listing page loads. 2.Product cards are displayed. 3.Page title matches the category.	Pass
22	PLP_017	Product card information	Verify each product card shows required information.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to any product listing page. 2.Observe individual product cards.	Any product listing page URL	1.Each card should display: product image, product title, price range, minimum order quantity (MOQ), and supplier name.	1.Product cards display image, title, price range, MOQ, and supplier name.	Pass
23	PLP_018	Sort by price	Verify products can be sorted by price.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to a product listing page. 2.Locate the 'Sort by' dropdown. 3.Select 'Price: Low to High'. 4.Observe product reordering.	Sort option: Price: Low to High	1.Products should be re-sorted with lowest priced items appearing first.	1.Products are sorted with lowest price first.	Pass
24	MODULE: PRODUCT DETAIL PAGE								
25	PDP_019	Navigate to product detail page	Clicking a product card should open the product detail page.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to https://www.albaba.com. 2.Search for 'Laptop'. 3.Click on any product card from the results.	Search keyword: 'Laptop'	1.User should be navigated to the product detail page. 2.Product name, images, price, and supplier info should be displayed.	1.Product detail page loads. 2.Product name, images, price, and supplier info are displayed.	Pass
26	PDP_020	Product image gallery navigation	Verify product images can be navigated on the detail page.	1.Application should be up and running. 2.Homepage should be loaded.	1.Navigate to a product detail page. 2.Observe the product image gallery. 3.Click next/previous arrows or thumbnails.	Any product detail page URL	1.Multiple images should be displayed. 2.Navigating arrows/thumbnails should cycle through images. 3.All images should load without errors.	1.Multiple images are present. 2.Navigation works correctly. 3.Images load without errors.	Pass
27	PDP_021	'Contact Supplier' button	Verify 'Contact Supplier' opens an inquiry form.	1.Application should be up and running. 2.User should be logged in.	1.Navigate to a product detail page. 2.Click 'Contact Supplier' button.	Logged-in session, Product detail page URL	1.An inquiry/message form should appear. 2.User should be able to compose and send a message to the supplier.	1.Inquiry form appears. 2.User can compose and send a message.	Pass

	A	B	C	D	E	F	G	H	I
1	Test id	Test Summary	Test Description	Precondition	Test Steps	Test Data	Expected Result	Actual Result	Status
28	MODULE: CART FUNCTIONALITY								
	CRT_022	Add product to cart	Verify a product can be added to the cart.	1.Application should be up and running. 2.User should be logged in.	1.Navigate to a product detail page. 2.Enter a valid quantity meeting the MOQ. 3.Click 'Add to Cart' or equivalent.	Logged-in session. Valid quantity (e.g., 50 units).	1.Product should be added to the cart. 2.Cart icon count in navigation should increment. 3.Confirmation message 'Item added to cart' should be displayed.	1.Product is added to cart. 2.Cart icon count increments. 3.Confirmation message is displayed.	Pass
29	CRT_023	View cart contents	Verify cart page displays added products correctly.	1.Application should be up and running. 2.User should be logged in. 3.At least one product should be in the cart.	1.Click the cart icon in the top navigation. 2.Observe the cart page contents.	Cart with at least one product.	1.Cart page should display all added products. 2.Each item should show product name, image, unit price, quantity, and subtotal.	1.Cart page displays all products. 2.Each item shows all required details.	Pass
30	CRT_024	Remove product from cart	Verify a product can be removed from the cart.	1.Application should be up and running. 2.User should be logged in. 3.At least one product should be in the cart.	1.Navigate to the cart page. 2.Click the 'Delete' or trash icon for a product. 3.Confirm removal.	Cart with at least one product.	1.Product should be removed. 2.Cart should reflect the updated item list. 3.Empty cart message should appear if no items remain.	1.Product is removed. 2.Cart is updated. 3.Empty cart message shown when cart is empty.	Pass
31	MODULE: CHECKOUT FLOW								
32	CHK_025	Initiate checkout from cart	Verify user can proceed to checkout from the cart.	1.Application should be up and running. 2.User should be logged in. 3.At least one product should be in the cart.	1.Navigate to the cart page. 2.Select product(s) to order. 3.Click 'Start Order' or 'Proceed to Checkout'.	Logged-in session. Cart with products.	1.User should be navigated to the checkout page. 2.Product details should be pre-filled.	1.User is navigated to checkout. 2.Product details are pre-filled.	Pass
33	CHK_026	Shipping address entry	Verify user can enter a shipping address during checkout.	1.Application should be up and running. 2.User should be logged in. 3.At least one product should be in the cart.	1.Proceed to checkout. 2.Navigate to the shipping address section. 3.Enter valid address details (name, address, city, country, postal code, phone). 4.Click 'Save'.	Valid shipping address details.	1.Shipping address should be saved. 2.Saved address should appear as the selected destination.	1.Shipping address is saved. 2.Address is selected as destination.	Pass
34	CHK_027	Order summary accuracy	Verify order summary displays accurate totals.	1.Application should be up and running. 2.User should be logged in. 3.At least one product should be in the cart.	1.Proceed to checkout. 2.Observe the order summary section.	Cart with known products and quantities.	1.Summary should display product name, quantity, unit price, and total. 2.Subtotal and grand total should be mathematically accurate.	1.Summary displays correct product details. 2.Totals are accurate.	Pass
35	CHK_028	Checkout without mandatory fields	Attempt to place order without filling mandatory fields.	1.Application should be up and running. 2.User should be logged in. 3.At least one product should be in the cart.	1.Proceed to checkout. 2.Skip the shipping address section. 3.Click 'Submit Order'.	Logged-in session. Incomplete form.	1.Order should not be placed. 2.Validation errors should highlight all mandatory empty fields.	1.Order is not placed. 2.Validation errors are displayed for mandatory fields.	Pass
36	MODULE: USER PROFILE / ACCOUNT SETTINGS								
37	ACC_029	View account profile	Verify user can view their account profile page.	1.Application should be up and running. 2.User should be logged in.	1.Login to Alibaba.com. 2.Hover over the account name in the top navigation. 3.Click 'My Alibaba' or 'Account'.	Valid logged-in user session.	1.Account profile page should load. 2.User's name, email, and account details should be displayed.	1.Account page loads. 2.User details are displayed.	Pass
38	ACC_030	Change account password	Verify user can change password from account settings.	1.Application should be up and running. 2.User should be logged in.	1.Navigate to Account Settings. 2.Locate 'Change Password'. 3.Enter current password. 4.Enter and confirm new password. 5.Click 'Save'.	Current valid password. New valid password (min 8 chars).	1.Password should be changed successfully. 2.'Password changed successfully' message should be displayed. 3.Login with new password should succeed.	1.Password is changed. 2.Success message is displayed. 3.Login with new password works.	Pass
39									